# Haochen Tian

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## Research Interests

**Scalable Physical AI** End-to-End Policy, Closed-Loop Training, Reinforcement Learning, World Model
**General MultiModal AI** Unified Foundation Model, Vision-Language Model, Alignment, Evaluation

## Education

**Institute of Automation, Chinese Academy of Sciences** Sep. 2023 - Present
*Ph.D.* in Computer Science & Technology, Advisor Prof. Tieniu Tan (IEEE Fellow)

**Wuhan University** Sep. 2019 - Jun. 2023
*B.S.* in Spatial Informatics & Digital Technology

## Selected Publications

∗ Equal Contribution † Project Lead ✉ Corresponding Author

### *Physical AI*

**SimScale: Learning to Drive via Real-World Simulation at Scale**
**Haochen Tian**, Tianyu Li, Haochen Liu, Jiazhi Yang, Guang Li, Junli Wang, Yinfeng Gao, Zhang Zhang, Liang Wang, Hangjun Ye, Tieniu Tan, Long Chen, Hongyang Li
**CVPR 2026 Oral (Top 0.88%, 141/16092).** Paper | Web | Code | Dataset | Oral Pre

**DriveZero: End-to-End Driving beyond Human Demonstrations**
**Haochen Tian as Core Contributor**, and Xiaomi L3 Team
Tech Report 2026. Paper | Web | Code

**WorldEngine: Towards the Era of Post-Training for Autonomous Driving**
OpenDriveLab, Huawei, and **Haochen Tian** as Contributor
Tech Report 2026. Paper | Web | Code

**OneVL: One-Step Latent Reasoning and Planning with Vision-Language Explanation**
Xiaomi Embodied Intelligence Team, and **Haochen Tian** as Contributor
Tech Report / **NeurIPS 2026**. Paper | Web | Code

**PlannerRFT: Reinforcing Diffusion Planners through Closed-Loop and Sample-Efficient Fine-Tuning**
Hongchen Li, Tianyu Li, Jiazhi Yang, **Haochen Tian**, Caojun Wang, Lei Shi, Mingyang Shang, Zengrong Lin, Gaoqiang Wu, Zhihui Hao, Xianpeng Lang, Jia Hu, Hongyang Li
**CVPR 2026.** Paper | Web

**Reinforced Refinement with Self-Aware Expansion for End-to-End Autonomous Driving**
Haochen Liu, Tianyu Li, Haohan Yang, Li Chen, Caojun Wang, Ke Guo, **Haochen Tian**, Hongchen Li, Hongyang Li, Chen Lv
**TPAMI 2026.** Paper

**DriveFine: Refining-Augmented Masked Diffusion VLA for Precise and Robust Driving**
Chenxu Dang, Sining Ang, Yongkang Li, **Haochen Tian**, Jie Wang, Guang Li, Hangjun Ye, Jie Ma, Long Chen, Yan Wang
**ECCV 2026.** Paper | Code

**SimVTA: Simulation-Grounded Video-Trajectory Alignment for Driving World Action Models**
Changhui Jing, **Haochen Tian**†, Hongchen Li, Tianyu Li, Caojun Wang, Zihao Zhang, Yanhao Liu, Chengen Xie, Bailin Li, Hao Zhao, Zhenjie Yang
Preprint 2026. Paper | Web| Code

### *MultiModal AI*

**A Survey of Unified Multimodal Understanding and Generation: Advances and Challenges**
Yan Yang*, **Haochen Tian***†, Yang Shi*, Wulin Xie*, Yi-Fan Zhang✉, Yuhao Dong, Yibo Hu, Liang Wang, Ran He, Caifeng Shan, Chaoyou Fu✉, Tieniu Tan
Preprint 2025. Paper | Web

**RealUnify: Do Unified Models Truly Benefit from Unification?**
**Haochen Tian** as Core Contributor.
**CVPR 2026.** Paper | Code | Dataset

**MME-RealWorld: Could Your MLLM Challenge High-Resolution Real-World Scenarios that are Difficult for Humans?**
Yi-Fan Zhang, Huanyu Zhang, **Haochen Tian**, Chaoyou Fu, Shuangqing Zhang, Junfei Wu, Feng Li, Kun Wang, Qingsong Wen, Zhang Zhang, Liang Wang, Rong Jin, Tieniu Tan
**ICLR 2025.** Paper | Web | Code | Dataset

**MM-RLHF: The Next Step Forward in Multimodal LLM Alignment**
Yi-Fan Zhang, Tao Yu, **Haochen Tian**, Chaoyou Fu, Peiyan Li, Jianshu Zeng, Wulin Xie, Yang Shi, Huanyu Zhang, Junkang Wu, Xue Wang, Yibo Hu, Bin Wen, Fan Yang, Zhang Zhang, Tingting Gao, Di Zhang, Liang Wang, Rong Jin, Tieniu Tan
**ICML 2025.** Paper | Web | Code | Dataset

**BaseReward: A Strong Baseline for Multimodal Reward Model**
Yi-Fan Zhang, Haihua Yang, Huanyu Zhang, Yang Shi, Zezhou Chen, **Haochen Tian**, Chaoyou Fu, Haotian Wang, Kai Wu, Bo Cui, Xu Wang, Jianfei Pan, Haotian Wang, Zhang Zhang, Liang Wang
**ICLR 2026.** Paper

## Open-sourced Projects

**AlpaSim: A Modular, Lightweight, and Data-Driven Research Simulator for Autonomous Driving**
NVIDIA, and **Haochen Tian** as Contributor
Code | Challenge

## Experience

**Xiaomi EV, Embodied Intelligence Team** — Sep. 2025 - Present
*Research Intern, **Top Talent Program*** — *Beijing, China*

- Worked with Prof. Long Chen and Naiyan Wang
- Worked on scalable closed-loop training for end-to-end autonomous driving.

**Shanghai AI Lab → The University of Hong Kong, OpenDriveLab** — Jun. 2024 - Present
*Research Intern* — *Hong Kong SAR, China*

- Worked with Prof. Hongyang Li
- Worked on human feedback reward for end-to-end autonomous driving.

**Megvii Research, Foundation Model Group** — Feb. 2024 - Mar. 2024
*Research Intern* — *Beijing, China*

- Worked on 3D-Aware Vision-Language-Action model for end-to-end autonomous driving.

## Honors / Awards

- **BYD Scholarship (Top 0.6%) (比亚迪奖学金)** 2026
- **Xiaomi Top Talent Program (小米顶尖人才计划)** 2026
- **CVPR Oral (Top 0.88%)** 2026
- Merit Student of University of Chinese Academy of Sciences 2024
- Outstanding Graduate of Wuhan University 2023
- **Wangzhizhuo Innovation Scholarship (Top 1%) (王之卓创新人才奖学金)** 2023

- **Gold Award of "Internet+" Innovation Competition** 2022
- Merit Student of Wuhan University 2021/2022

## Academic Service

**Competition Organizer**
AlpaSim End-to-End Closed-Loop Challenge 2026

**Workshop Organizer**
CVPR 2026 Workshop on From Labs to Life: Embodied Intelligence in the Wild

**Standards Working Group Member**
IEEE Standard P3474: Human Intentions and AI Alignment in Autonomous Driving Agents

**Conference/Journal Reviewer**
CVPR, ECCV, NeurIPS, ICLR, ICRA, RSS, WACV, BMCV, RA-L, T-PAMI, etc.

## Invited Talks

**Scalable Real-World Simulation Reshapes End-to-End Policy Learning** 2025
AIR@THU, SJTU, 机器之心, 自动驾驶之心, 3D 视觉工坊, 计算视觉 Life, 青稞社区

**DriveZero: End-to-End Driving beyond Human Demonstrations** 2026.10
Valeo.ai

## Skills

**Language**
Chinese (Native) | English (Proficient CET6 500+, CET4 600+)

**Programming**
Python, C, C++, Matlab, Java, JavaScript, SQL

**Frameworks/Tools**
PyTorch, Transformers, OpenMMLab, Git, LaTeX, Linux, Docker